

## Week 1 Assessment Event: Pump Readiness Closeout

Values, state, type evidence, boundaries, and Boolean readiness

Student copy. Show evidence. Final answers without trace, test, or debug reasoning may not show full mastery.

|       |          |         |                |                                                      |  |
|-------|----------|---------|----------------|------------------------------------------------------|--|
| Name  |          | Section |                | Date                                                 |  |
| Score | _____/50 |         | Mastery signal | secure / mostly secure / developing / support needed |  |

### Mission

The Week 1 pump record is ready for closeout only if the numeric flow evidence, pressure evidence, approval label, setup evidence, and retry evidence all support readiness. Your task is to read Python evidence the way an engineer would read a lab record: trace current values, evaluate exact boundaries, test claims, debug one unsafe statement, and write a closeout decision.

### Allowed Python tools for this event

Assignments, arithmetic expressions, `float(...)`, strings, Booleans, comparisons, `and`, `or`, `not`, and `print(...)`. Do not write a function, loop, or `if/else` solution.

### Point map Manage your time

| Points | Section                       | Evidence focus    |
|--------|-------------------------------|-------------------|
| 8      | Output and state evidence     | predict; trace    |
| 8      | Types, casts, and truthiness  | recognize; debug  |
| 12     | Boundary and Boolean trace    | predict; explain  |
| 8      | Test-case reasoning           | test; explain     |
| 8      | Completion and debugging      | implement; debug  |
| 6      | Scenario closeout explanation | transfer; explain |

### Reference code Use for all questions unless a question changes one value

[trace](#)[predict](#)

```
1 target_lpm = 21.0
2 tolerance_text = "0.60"
3 low_limit_lpm = target_lpm - float(tolerance_text)
4 high_limit_lpm = target_lpm + float(tolerance_text)
5
6 flow_text = "21.60"
7 flow_lpm = float(flow_text)
8 pressure_psi = 43
9 pressure_limit_psi = 45
10 operator_initials = ""
11 approval_label = "reviewed"
12 sensor_ready = True
13 retry_count = 0
14
15 flow_low_ok = flow_lpm >= low_limit_lpm
16 flow_high_ok = flow_lpm <= high_limit_lpm
17 flow_ok = flow_low_ok and flow_high_ok
18 pressure_ok = pressure_psi <= pressure_limit_psi
19 approval_ok = approval_label == "reviewed"
20 setup_ok = sensor_ready and operator_initials != ""
21 retry_ok = retry_count == 0
22 ready_to_close = flow_ok and pressure_ok and approval_ok and setup_ok and retry_ok
23
24 print("limits", low_limit_lpm, high_limit_lpm)
25 print("flow", flow_lpm)
26 print("flow ok", flow_ok)
27 print("setup ok", setup_ok)
28 print("close", ready_to_close)
```

**A** Output and state evidence

predict

trace

| Pts | Prompt                                                                                                | My response |
|-----|-------------------------------------------------------------------------------------------------------|-------------|
| 2   | What numeric values are stored in <code>low_limit_lpm</code> and <code>high_limit_lpm</code> ?        |             |
| 2   | What exactly is displayed by the first two <code>print(...)</code> lines?                             |             |
| 2   | What values are displayed for <code>flow ok</code> , <code>setup ok</code> , and <code>close</code> ? |             |
| 2   | Which one line makes <code>setup_ok</code> false even though <code>sensor_ready</code> is true?       |             |

**B** Types, casts, and truthiness

recognize

debug

| Pts | Prompt                                                                                                                               | My response |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 2   | After line 6 runs, what are the value and type of <code>flow_lpm</code> ?                                                            |             |
| 2   | After line 6 runs, what are the value and type of <code>flow_text</code> ? Explain whether <code>float(flow_text)</code> changed it. |             |
| 2   | If Python evaluated <code>flow_text + tolerance_text</code> , what value would that expression produce? Why?                         |             |
| 2   | Why is <code>bool(approval_label)</code> not enough evidence that the approval label is correct?                                     |             |

**C** Boundary and Boolean trace

predict

explain

| Pts | Name/expression                                                | Value | Reason |
|-----|----------------------------------------------------------------|-------|--------|
| 2   | flow_low_ok                                                    |       |        |
| 2   | flow_high_ok                                                   |       |        |
| 2   | flow_ok                                                        |       |        |
| 2   | pressure_ok                                                    |       |        |
| 2   | ready_to_close                                                 |       |        |
| 2   | Which required condition controls the final closeout decision? |       |        |

**Exam-style check**

Circle the one expression that is **not** equivalent to the intended readiness rule.

- A. flow\_ok and pressure\_ok and approval\_ok and setup\_ok and retry\_ok
- B. ready\_to\_close
- C. (flow\_ok and pressure\_ok) and (approval\_ok and setup\_ok) and retry\_ok
- D. flow\_ok or pressure\_ok or approval\_ok or setup\_ok or retry\_ok

Explain your choice briefly: \_\_\_\_\_

**D** Test-case reasoning

test

explain

| Pts | Prompt                                                                                          | My response |
|-----|-------------------------------------------------------------------------------------------------|-------------|
| 2   | Give a flow value exactly at the upper boundary. State the expected value of flow_high_ok.      |             |
| 2   | Give a flow value just above the upper boundary. State which Boolean check should become false. |             |
| 2   | Give a setup case that should make setup_ok true without changing flow or pressure.             |             |
| 2   | Which one test would most directly catch a mistaken use of or in ready_to_close?                |             |

## E Completion and debugging

[implement](#)[debug](#)

Complete the expression below so it matches the all-required readiness rule. Use only the existing Boolean variables.

`ready_to_close = -----`

| Pts | Prompt                                                                                                                                                              | My response |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 3   | Write the completed expression.                                                                                                                                     |             |
| 2   | A teammate writes <code>ready_to_close = flow_ok or approval_ok</code> . Why is that unsafe?                                                                        |             |
| 3   | A teammate writes <code>approval_ok = bool(approval_label)</code> . Give one approval-label value that would make this check pass but should not count as reviewed. |             |

## F Scenario closeout decision

[transfer](#)[explain](#)

Write the Week 1 closeout note that should go in the pump record. It must name the final readiness decision, cite at least two pieces of Python evidence, and identify the next action before the record can close.

| Pts | Closeout note                                                                                                                                                                                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   |                                                                                                                                                                                                                         |
| 2   | If your answer shows a gap, circle the most useful support route: state tracing / type and cast evidence / boundary comparisons / Boolean connectors / testing / written explanation. Name one next small action: _____ |

### Mastery interpretation

Secure work shows current state, type/cast evidence, exact-boundary reasoning, Boolean connector control, a useful test, and a correct engineering closeout decision. If one of those evidence types is missing, use the tag on that section to choose the next support step.

Scratch and trace space Use this page for calculations, state tables, or output notes

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